1) Enlist objectives of software testing.

: >To detect Errors
>To meet specified requirements
>To Improve Quality
>To Reduce risk of failures
>To ensure Reliability

2) Define following terms:

: i) Error: A mistake made by a Developer that may cause the program to work incorrectly.

ii) Fault: A Fault is an incorrect step, process or data definition in a Computer Program that may cause a failure.

iii) Bug : A Bug is a fault in a Computer program or a system which can cause it to behave incorrectly or generate incorrect result.

iv) Failure: A Failure is inability of a system or a component to do its required function within specified Limits.

3) Define Driver and Stub.

: i) Driver: A Driver is test code which is used call or control a software module which is being module, especially when higher level modules are not developed yet.

ii) Stub: A Stub is a dummy or temporary module used to replace lower level modules while testing when they’re not developed.

4) Write important six test cases for the “Login Form” of the Facebook website

Test Case ID	Test Scenario	Input Data	Expected Result
TC01	Login with valid username and valid password	Valid email & valid password	User should be logged in successfully
TC02	Login with valid username and invalid password	Valid email & invalid password	Error message should be displayed
TC03	Login with invalid username and valid password	Invalid email & valid password	Error message should be displayed
TC04	Login with both fields blank	Username = blank, Password = blank	Validation message should be shown
TC05	Login with password field blank	Valid email, Password = blank	Prompt to enter password
TC06	Login with username/email field blank	Username = blank, Valid password	Prompt to enter username/email

5) Differentiate between Smoke Testing and Sanity Testing (any four points).

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Smoke Testing	Sanity Testing
To check the critical functionality of a program is working fine.	To check the new functionality or after bugs have been fixed.
Objective is to verify the stability	Objective is to verify the rationality
Performed by developers and testers	Performed generally by testers
Usually documented or scripted	Not Documented or scripted.

6) Write steps to prepare test cases.

: i) Understand the requirement of the app/software
ii) Identify the test cases
iii) Design the test cases with input/output data
iv) Review and refine the test cases
v) Assign test case ids and priorities
vi) STore the test cases for future testing

7)List any four skills of software tester.

: i) Analytical skills
ii) Attention to detail
iii) Knowledge of testing tools
iv) Commmunication skills

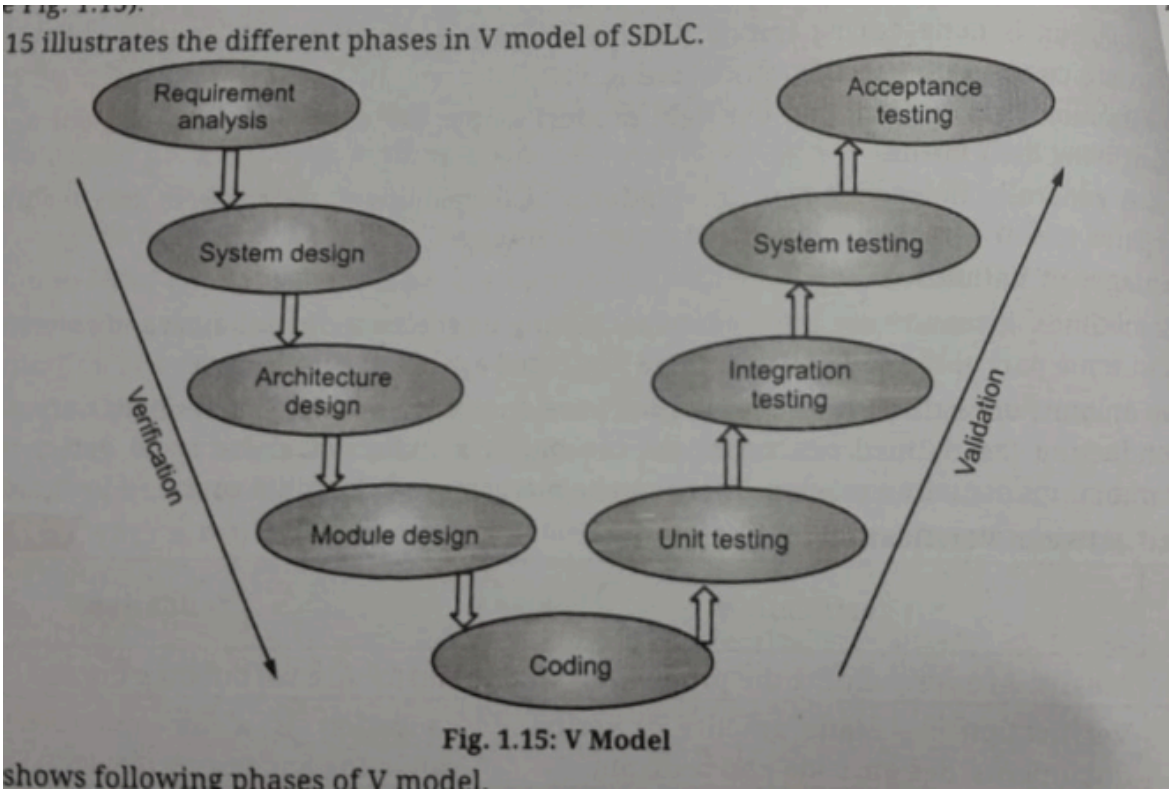
8)Describe entry criteria and exit criteria of software testing

: i) Entry criteria: Conditions that must be met before testing can begin
Ex: Test plan and cases should be prepared, software build is stable, required test environment is setup
ii) Exit criteria : Conditions that must be met before concising the test
Ex: All planned tests are executed, all critical or major defects are fixed, test cases reports are prepared and approved.

9)Define the terms error and defect in relation with software testing.

: i) Error: An error is a mistake by the developer that can cause the program to behave incorrectly or generate incorrect results
ii) Defect: A defect is a flaw caused by an error in a program that may cause it to fail or behave in unexpected ways.

10) Describe V-model with labelled diagram.



The V Model is a Software Development and Testing Module where each Development phase is directly associated with corresponding testing phases. It is structured like a V where Development Phases are on right and Testing are on the left.

i) V Model is a Software Development model which emphasizes verification and validation.
ii)The development phases are on the left and the testing phases are on the right.
iii)Each development phase has a directly related testing phase
iv)Helps in early defect detection and better software quality.

11) Differentiate between Alpha Testing and Beta Testing.
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Alpha Testing	Beta Testing
Performed By developers	Performed by Actual users
Done Before releasing	Done after Alpha
Done in controlled environment	Done in the actual user site
Testers are the developers or QA Team	Testers are the actual users
Focuses on FInding all possible bugs	Focuses on real usability
Helps in improving sf quality before release	Helps in validating user satisfaction

12) Describe acceptance testing with its advantages.
: Acceptance testing is the final phase of software testing carried out to determine whether the system meets the business and user requirements. It is performed from the end-user's perspective to ensure the software is ready for deployment.

i)Ensures the software **fulfills all customer requirements**.
ii)Helps **reduce the risk of failures** when the system is used in a real environment.
iii)Builds **confidence for both users and developers** about software quality.
iv)Identifies **missing functionality, errors, or usability issues** before release.

13)State the process of performance testing.
: Identify performance requirements – Determine response time, throughput, and load limits.

Plan and design test cases – Decide test scenarios, workloads, and performance metrics.

Set up the test environment – Configure hardware, software, tools, and network settings.

Execute performance tests – Run load, stress, and endurance tests.

Monitor and analyze results – Measure system behavior and identify bottlenecks.

Tune and retest – Optimize the system and re-test to ensure performance goals are met.

14)Difference between Verification and Validation.

Verification	Validation
Verifies whether the product is built correctly	Verifies whether the right product is built
Focus on process and documentation	Focus on actual software development
Does not involve code execution	Involve code execution
During Development phase	During testing phase
Includes reviews , inspections and walkthroughs	Includes testing activites
Detects early defects	Detects defects int he final prodcut